

How gradient noise becomes generator bias in multilevel Picard, and what a certified gradient retraction does: a one-step analysis

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Abstract

We analyse a single generator evaluation inside a multilevel Picard (MLP) recursion for semilinear parabolic PDEs whose solution has ridge structure. For the centred terminal-block gradient estimator we show that the estimation noise orthogonal to the ridge direction has trace exactly $(d-1)c_h/M$, with c_h independent of the dimension d (Lemma 1). Passed through a quadratic generator, this noise produces a deterministic bias equal to $-\frac{\lambda}{2}(d-1)c_h/M$ (Proposition 2); through a norm generator it produces a bias of order $\sqrt{d/M}$ (Proposition 3). Retracting the gradient onto a PDE-certified set before the generator bounds the generator error by a d -independent constant, and when the set lies in the ridge direction it makes the law of the corrected generator value exactly independent of d (Proposition 5). For generators that are linear in the gradient along the ridge direction, the orthogonal noise never reaches the generator (Proposition 8), which explains why the certified retraction does not improve equal-cost accuracy in that class. The analysis concerns one generator call (equivalently one Picard level); how deeper levels amplify the effect is documented empirically, not proved here.

1 Setting

Consider $\partial_t u + \mathcal{L}u + f(t, x, u, z) = 0$, $u(T, \cdot) = g$, with $z = \sigma \nabla_x u$ for a scalar $\sigma > 0$ and $\mathcal{L}u = b \cdot \nabla u + \frac{\sigma^2}{2} \Delta u$ for a constant drift $b \in \mathbb{R}^d$. Fix a unit vector $w \in \mathbb{R}^d$, write $P_\perp = I - ww^\top$, and use the following assumptions.

Assumption 1 (Ridge structure). (i) $g(x) = \varphi(w \cdot x)$ with $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ Lipschitz with constant L_φ ; (ii) $b = \beta w$ for some $\beta \in \mathbb{R}$; (iii) the exact solution is a ridge function, $u^*(t, x) = \psi(t, w \cdot x)$, so that $z^*(t, x) = \sigma \partial_s \psi(t, w \cdot x) w$.

Assumption 1(iii) holds whenever f depends on z only through $\|z\|$ or through $w \cdot z$ and the PDE has a unique solution in the relevant class (translation invariance in the directions orthogonal to w). It covers the ridge log-sum-exp HJB (P1), the norm-driver HJB (P4), the counterexample of Hutzenthaler and Nguyen [1] ($w = e_1$), and the viscous-Burgers family $u^* = \text{sigmoid}(at + \sum_i x_i)$ (P2/P3, $w = d^{-1/2} \mathbf{1}$).

The estimator. At a child point (τ, x) with $h = T - \tau > 0$, MLP estimates the gradient by Monte Carlo. We analyse the centred terminal block

$$\hat{z} = \frac{1}{M} \sum_{k=1}^M \Delta_k \frac{G_k}{\sqrt{h}}, \quad \Delta_k = g(x + \beta h w + \sigma \sqrt{h} G_k) - g(x), \quad G_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_d). \quad (1)$$

This is the complete child gradient at the first Picard level and the terminal contribution at every deeper level. It is also the noise source used in the lower-bound argument of [1]. Write $s = w \cdot x$, $\xi_k = w \cdot G_k$, $G_{\perp,k} = P_{\perp} G_k$, so that $\Delta_k = \varphi(s + \beta h + \sigma \sqrt{h} \xi_k) - \varphi(s)$ depends on G_k only through ξ_k . Define

$$\widehat{z}_{\parallel} = w \cdot \widehat{z}, \quad \widehat{z}_{\perp} = P_{\perp} \widehat{z}, \quad Q_M = \frac{1}{Mh} \sum_{k=1}^M \Delta_k^2, \quad c_h(s) = \mathbb{E} Q_M = \frac{\mathbb{E} \Delta_1^2}{h}.$$

2 The orthogonal noise

Lemma 1 (Orthogonal noise is isotropic, mean zero, and scales with $d-1$). *Under Assumption 1(i)–(ii):*

- (i) $\mathbb{E} \widehat{z}_{\perp} = 0$ and $\mathbb{E} \widehat{z} = (\mathbb{E} \widehat{z}_{\parallel}) w$;
- (ii) the law of $\widehat{z}_{\parallel}$ and of Q_M does not depend on d ;
- (iii) conditionally on $(\xi_1, \dots, \xi_M)$, $\widehat{z}_{\perp} \sim \mathcal{N}(0, \frac{Q_M}{M} P_{\perp})$;
- (iv) $\mathbb{E} \|\widehat{z}_{\perp}\|^2 = (d-1) c_h(s)/M$, and $c_h(s) \leq 2L_{\varphi}^2(\sigma^2 + \beta^2 h)$;
- (v) $\mathbb{E} \|\widehat{z}_{\perp}\| = \mathbb{E}[\sqrt{Q_M}] \mathbb{E} \chi_{d-1}/\sqrt{M} \geq \mathbb{E}[\sqrt{Q_M}] \frac{d-1}{\sqrt{(d+1)M}}$, where χ_{d-1} denotes a chi random variable with $d-1$ degrees of freedom.

Proof. Because G_k is standard Gaussian and $w \perp \text{range } P_{\perp}$, the scalar ξ_k and the vector $G_{\perp,k} \sim \mathcal{N}(0, P_{\perp})$ are independent, and the pairs are independent across k . Since Δ_k is a function of ξ_k , $\widehat{z}_{\parallel} = (M\sqrt{h})^{-1} \sum_k \Delta_k \xi_k$ and Q_M are functions of $(\xi_1, \dots, \xi_M)$, whose law does not involve d ; this gives (ii). Conditionally on the ξ 's, $\widehat{z}_{\perp} = (M\sqrt{h})^{-1} \sum_k \Delta_k G_{\perp,k}$ is a linear combination of independent $\mathcal{N}(0, P_{\perp})$ vectors with fixed coefficients, hence Gaussian with mean zero and covariance $(M^2 h)^{-1} \sum_k \Delta_k^2 P_{\perp} = (Q_M/M) P_{\perp}$; this gives (iii) and the first part of (i). The second part of (i) follows from $\widehat{z} = \widehat{z}_{\parallel} w + \widehat{z}_{\perp}$. For (iv), $\mathbb{E}[\|\widehat{z}_{\perp}\|^2 \mid \xi] = \text{tr}((Q_M/M) P_{\perp}) = (d-1) Q_M/M$; take expectations. The bound on c_h follows from $|\Delta_k| \leq L_{\varphi}(|\beta|h + \sigma\sqrt{h}|\xi_k|)$ and $(a+b)^2 \leq 2a^2 + 2b^2$. For (v), conditionally on ξ , $\|\widehat{z}_{\perp}\| = \sqrt{Q_M/M} \chi_{d-1}$ with χ_{d-1} independent of ξ . For a nonnegative random variable X , Hölder's inequality with exponents $3/2$ and 3 applied to $X^2 = X^{2/3} X^{4/3}$ gives $\mathbb{E} X^2 \leq (\mathbb{E} X)^{2/3} (\mathbb{E} X^4)^{1/3}$, that is $\mathbb{E} X \geq (\mathbb{E} X^2)^{3/2} (\mathbb{E} X^4)^{-1/2}$. With $\mathbb{E} \chi_k^2 = k$ and $\mathbb{E} \chi_k^4 = k(k+2)$ this yields $\mathbb{E} \chi_k \geq k/\sqrt{k+2}$; take $k = d-1$. $\square$

Under Assumption 1(iii) the exact gradient has no orthogonal component, $P_{\perp} z^* = 0$. Every unit of $\mathbb{E} \|\widehat{z}_{\perp}\|^2$ is therefore pure estimation noise, and it grows linearly in d at fixed M .

3 What a curved generator does with this noise

Proposition 2 (Quadratic generator: exact Jensen bias). *Let $f(z) = -\frac{\lambda}{2} \|z\|^2$, $\lambda > 0$. Under Assumption 1(i)–(ii),*

$$\mathbb{E} f(\widehat{z}) - f(\mathbb{E} \widehat{z}) = -\frac{\lambda}{2} \left(\text{Var}(\widehat{z}_{\parallel}) + \frac{(d-1) c_h(s)}{M} \right).$$

If in addition Assumption 1(iii) holds, then $\mathbb{E} f(\widehat{z}) - f(z^) = A_M(s) - \frac{\lambda}{2} (d-1) c_h(s)/M$, where $A_M(s) = f(\mathbb{E} \widehat{z}) - f(z^*) - \frac{\lambda}{2} \text{Var}(\widehat{z}_{\parallel})$ does not depend on d .*

Proof. Since $\hat{z}_\perp \perp w$, $\|\hat{z}\|^2 = \hat{z}_\parallel^2 + \|\hat{z}_\perp\|^2$, so $\mathbb{E}f(\hat{z}) = -\frac{\lambda}{2}((\mathbb{E}\hat{z}_\parallel)^2 + \text{Var}(\hat{z}_\parallel) + \mathbb{E}\|\hat{z}_\perp\|^2)$. By Lemma 1(i), $f(\mathbb{E}\hat{z}) = -\frac{\lambda}{2}(\mathbb{E}\hat{z}_\parallel)^2$, and (iv) gives the result. Under (iii), $z^\star = (w \cdot z^\star)w$ and $w \cdot z^\star$ depends only on (τ, s) ; together with Lemma 1(ii) this makes A_M independent of d . $\square$

Proposition 3 (Norm generator: bias of order $\sqrt{d/M}$). *Let $f(z) = -\lambda\|z\|$, $\lambda > 0$. Under Assumption 1(i)–(ii),*

$$\mathbb{E}[\sqrt{Q_M}] \frac{d-1}{\sqrt{(d+1)M}} \leq \mathbb{E}\|\hat{z}\| \leq \sqrt{(\mathbb{E}\hat{z}_\parallel)^2 + \text{Var}(\hat{z}_\parallel) + \frac{(d-1)c_h(s)}{M}}.$$

Consequently, under Assumption 1(iii), $f(z^\star) - \mathbb{E}f(\hat{z}) \geq \lambda\left(\mathbb{E}[\sqrt{Q_M}] \frac{d-1}{\sqrt{(d+1)M}} - \|z^\star\|\right)$, where $\mathbb{E}\sqrt{Q_M}$ and $\|z^\star\|$ do not depend on d .

Proof. The lower bound uses $\|\hat{z}\| \geq \|\hat{z}_\perp\|$ and Lemma 1(v); the upper bound is Jensen’s inequality $\mathbb{E}\|\hat{z}\| \leq (\mathbb{E}\|\hat{z}\|^2)^{1/2}$ with the decomposition used in Proposition 2. $\square$

Remark 4 (The Hutzenthaler–Nguyen counterexample). In [1], $f_d(z) = \|P_\perp z\|$ with $w = e_1$, $\sigma = 1$, $\beta = 0$ and $g(x) = |x_1|$. Then $f_d(\hat{z}) = \|\hat{z}_\perp\|$ exactly, so $\mathbb{E}f_d(\hat{z})^2 = (d-1)c_h/M$ and $\mathbb{E}f_d(\hat{z}) \geq \mathbb{E}[\sqrt{Q_M}] (d-1)/\sqrt{(d+1)M}$, while $f_d(z^\star) = 0$. At the origin $c_h = \mathbb{E}[(\sqrt{h}|\xi|)^2]/h = 1$, which recovers the identity $\mathbb{E}f_d(V_N)^2 = (d-1)/N$ used in Appendix B of the manuscript. The counterexample is thus the special case in which the generator sees *only* the orthogonal noise.

4 What a certified retraction does

Let $\mathcal{C}(\tau, x) \subseteq \mathbb{R}^d$ be closed and convex with $z^\star(\tau, x) \in \mathcal{C}(\tau, x)$ (a valid certificate), and let f be L_C -Lipschitz on $\mathcal{C}$.

Proposition 5 (Certified retraction caps the generator error). *(a) For every realisation, $|f(\Pi_{\mathcal{C}}\hat{z}) - f(z^\star)| \leq L_C \min(\|\hat{z} - z^\star\|, \text{diam } \mathcal{C})$.*

(b) If $\mathcal{C} \subseteq \text{span}(w)$, write $\mathcal{C} = \{cw : c \in I\}$ for a closed interval I and let π_I be the projection onto I . Then $\Pi_{\mathcal{C}}\hat{z} = \pi_I(\hat{z}_\parallel)w$, so $\|\Pi_{\mathcal{C}}\hat{z} - z^\star\| = |\pi_I(\hat{z}_\parallel) - w \cdot z^\star| \leq |\hat{z}_\parallel - w \cdot z^\star|$. Under Assumption 1, if I depends only on $(\tau, w \cdot x)$ and $f(cw) = F(c)$ for a function F that does not depend on d , the law of $f(\Pi_{\mathcal{C}}\hat{z})$ does not depend on d .

Proof. (a) $z^\star = \Pi_{\mathcal{C}}z^\star$, both projections lie in $\mathcal{C}$, and metric projection onto a closed convex set is nonexpansive; also $\|\Pi_{\mathcal{C}}\hat{z} - z^\star\| \leq \text{diam } \mathcal{C}$. (b) For $y \in \mathbb{R}^d$ and $c \in \mathcal{C} \subseteq \text{span}(w)$, $\|y - c\|^2 = \|P_\perp y\|^2 + \|(w \cdot y)w - c\|^2$, so the minimiser over $\mathcal{C}$ depends on y only through $w \cdot y$; for $y = \hat{z}$ it is $\pi_I(\hat{z}_\parallel)w$. The law statement follows from Lemma 1(ii). $\square$

Remark 6 (Which component of the certificate does what). Part (b) shows that the subspace component removes the d -dependence: the orthogonal noise is discarded exactly. Part (a) shows that the magnitude component bounds the generator error by $L_C \text{diam } \mathcal{C}$. Both are needed in a recursion with an unbounded generator. Projecting onto $\text{span}(w)$ alone removes the d -growth but leaves the heavy tails of $\hat{z}_\parallel^2$ to be amplified at deeper levels; this matches the round-1 ablation on P1, where the span-only correction has mean skill 573 (median 2.35) while the box certificate has 0.17. For P1 the segment, its box hull and the ball all have diameters independent of d , so (a) gives a d -independent bound for each of them.

Corollary 7 (One averaged nonlinear correction). *Fix a parent (t, x) . Let child times $\tau_i \in [t, T]$ be i.i.d., child positions $X_i = x + \beta(\tau_i - t)w + \sigma W_{\tau_i - t}$, and let $\hat{z}_i$ be the estimator (1) at (τ_i, X_i) with fresh samples. For nonnegative weights $a(\tau)$ consider $\hat{C} = K^{-1} \sum_{i=1}^K a(\tau_i) f(\hat{z}_i)$ and $\hat{C}_C = K^{-1} \sum_{i=1}^K a(\tau_i) f(\Pi_C \hat{z}_i)$. Under Assumption 1, with $C^* = \mathbb{E}[a(\tau) f(z^*(\tau, X))]$:*

- (i) *for $f = -\frac{\lambda}{2} \|z\|^2$, $\mathbb{E}\hat{C} - C^* = B_0 - \frac{\lambda}{2} (d - 1) \mathbb{E}[a(\tau) c_{T-\tau}(w \cdot X)]/M$ with B_0 independent of d ;*
- (ii) *in the setting of Proposition 5(b), $|\mathbb{E}\hat{C}_C - C^*| \leq L_C \mathbb{E}[a(\tau) \min(|\hat{z}_\parallel - w \cdot z^*|, |I|)]$, a quantity independent of d .*

Proof. The law of $w \cdot X_i = s + \beta(\tau_i - t) + \sigma w \cdot W_{\tau_i - t}$ does not depend on d . Condition on (τ_i, X_i) and apply Proposition 2 for (i) and Proposition 5 for (ii). $\square$

5 The boundary: generators linear in the gradient along the ridge

Proposition 8 (Orthogonal noise is invisible to ridge-linear generators). *Let $f(u, z) = A(u) + B(u)(v \cdot z)$ with $v \in \text{span}(w)$. Then $f(\hat{u}, \hat{z}) = A(\hat{u}) + B(\hat{u})(v \cdot w) \hat{z}_\parallel$ for any value estimate $\hat{u}$; in particular $f(\hat{u}, \hat{z})$ does not depend on $\hat{z}_\perp$, and conditionally on $\hat{u}$ the generator is affine in $\hat{z}$, so the gradient noise produces no Jensen gap. Any remaining bias enters through the dependence between $\hat{u}$ and $\hat{z}_\parallel$.*

Proof. $v \cdot \hat{z}_\perp = 0$ because $v \in \text{span}(w)$ and $\hat{z}_\perp \perp w$. $\square$

The viscous-Burgers family has $f(u, z) = \sigma u \sum_i z_i = \sigma \sqrt{d} u (w \cdot z)$ with $w = d^{-1/2} \mathbf{1}$, so Proposition 8 applies. This is consistent with three empirical findings: the certified box does not win at equal cost on P2/P3, its benefit is confined to deep and under-sampled configurations, and the oracle decomposition attributes part of the P2 error to the correlation between u - and z -noise rather than to the gradient channel alone.

6 Scope

- The results are exact statements about one generator evaluation (and one averaged correction) with the terminal-block estimator. At deeper Picard levels the child gradient also contains nonlinear-correction terms; we do not prove how these propagate. The empirical oracle decomposition and the d -sweeps in the mechanism suites are the evidence for the recursive picture.
- The ridge assumption is what makes the statements exact. It is also the regime in which nonlinearity matters in high dimension for our benchmarks: when the terminal condition depends on many directions, concentration of measure makes the nonlinear correction nearly independent of x (Rosenbrock HJB, multi-direction log-sum-exp).
- All statements about the retraction require a valid certificate. Retractions that are not the identity at z^* (illegal tighter sets, shrinkage towards a centre) change the generator at the solution and are not covered.

7 Numerical check

The script `experiments/theory_onestep/verify_onestep_theory.py` samples (1) by brute force in $\mathbb{R}^d$ (it does not use the conditional law of Lemma 1) at $s = 0.3$, $h = 0.2$, for the P1 terminal $\varphi(s) = -\log \frac{1}{3} \sum_k e^{\lambda_k s}$, $\lambda = (0.5, 1.5, 3)$, $\sigma = \sqrt{2}$, $f = -\frac{1}{2}\|z\|^2$, certificate $I = [-3\sqrt{2}, -0.5\sqrt{2}]$, and for the P4 terminal $\varphi(s) = \beta^{-1} \log \cosh(\beta s)$, $\beta = 2$, $\sigma = \sqrt{2}$, $f = -\|z\|/\sqrt{2}$, certificate $I = [-\sqrt{2}, \sqrt{2}]$. It uses 20,000 replicates for $d \leq 200$ and 6,000 for $d = 1000$. Full output, including $M = 4$, is in `results/theory_onestep/onestep_theory_check.md`. The “certified” columns use the projected generator minus f evaluated at the projected mean gradient of the linear problem, a common reference across d .

Table 1: Monte Carlo check at $M = 16$. Quadratic rows: Jensen gap $\mathbb{E}f(\hat{z}) - f(\mathbb{E}\hat{z})$; norm rows: $\mathbb{E}\|\hat{z}\|$ with the bounds of Proposition 3. Generator bias is measured against the common reference described in the text.

driver	d	$\mathbb{E}\ \hat{z}_\perp\ ^2$		Jensen gap or $\mathbb{E}\ \hat{z}\ $		raw bias	certified	
		MC	theory	MC	theory / bounds		bias	median
quadratic	10	4.53	4.54	-2.81	-2.81	-2.81	-0.225	0.404
quadratic	50	24.84	24.70	-12.95	-12.88	-12.98	-0.245	0.347
quadratic	200	100.61	100.33	-50.84	-50.70	-50.85	-0.231	0.373
quadratic	1000	501.66	503.66	-251.34	-252.34	-251.31	-0.199	0.397
norm	10	0.49	0.48	0.845	[0.607, 0.922]	-0.29	-0.037	0.011
norm	50	2.62	2.61	1.642	[1.529, 1.725]	-0.86	-0.032	0.014
norm	200	10.69	10.62	3.195	[3.137, 3.314]	-1.95	-0.035	0.013
norm	1000	53.31	53.31	7.058	[7.029, 7.326]	-4.69	-0.032	0.012

In every row the orthogonal trace and the quadratic Jensen gap agree with Lemma 1(iv) and Proposition 2 within Monte Carlo error, the norm expectation lies between the bounds of Proposition 3, the raw generator bias grows linearly (quadratic) or like $\sqrt{d}$ (norm), and the distribution of the certified generator value is unchanged from $d = 10$ to $d = 1000$, as Proposition 5(b) predicts.

References

- [1] M. Hutzenthaler and T. A. Nguyen. Full history recursive multilevel Picard approximations suffer from the curse of dimensionality for the Hamilton–Jacobi–Bellman equation of a stochastic control problem. arXiv:2506.23969, 2025.